

Cotreatment With Clofazimine and Rapamycin Eliminates Drug-Resistant Tuberculosis by Inducing Polyfunctional Central Memory T-Cell Responses

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Mycobacterium tuberculosis, the causative agent of tuberculosis, is acquiring drug resistance at a faster rate than the discovery of new antibiotics. Therefore, alternate therapies that can limit the drug resistance and disease recurrence are urgently needed. Emerging evidence indicates that combined treatment with antibiotics and an immunomodulator provides superior treatment efficacy. Clofazimine (CFZ) enhances the generation of T central memory (T_{CM}) cells by blocking the $Kv1.3^+$ potassium channels. Rapamycin (RAPA) facilitates *M. tuberculosis* clearance by inducing autophagy. In this study, we observed that cotreatment with CFZ and RAPA potently eliminates both multiple and extensively drug-resistant (MDR and XDR) clinical isolates of *M. tuberculosis* in a mouse model by inducing robust T-cell memory and polyfunctional T_{CM} responses. Furthermore, cotreatment reduces the expression of latency-associated genes of *M. tuberculosis* in human macrophages. Therefore, CFZ and RAPA cotherapy holds promise for treating patients infected with MDR and XDR strains of *M. tuberculosis*.

Keywords. central memory T cells; clofazimine; extensively drug-resistant tuberculosis; helper T cells; immunotherapy; multidrug-resistance; rapamycin; tuberculosis.

Tuberculosis remains one of the biggest killers worldwide, claiming approximately 2 million lives and causing 9 million new infections every year [1]. Tuberculosis therapy, referred to as directly observed treatment short-course (DOTS), consists of multiple antibiotics that need to be administered for long durations and have severe side effects, which lead to non-compliance and the subsequent risk for the generation of multiple and extremely drug-resistant (MDR and XDR) organisms [2–4]. *M. tuberculosis* is acquiring drug resistance at a faster rate than the discovery of new classes of antibiotics, and totally drug-resistant (TDR) *M. tuberculosis* strains have already emerged [5,6]. Furthermore, these antibiotics are associated with substantial immune toxicity, which enhances host susceptibility to disease reactivation and reinfection [7,8].

Recently, we have reported that antimycobacterial antibiotics such as isoniazid (INH) partially eliminate antigen-

responding $CD4^+$ T cells, which makes animals vulnerable to disease reactivation and reinfection [9]. Furthermore, the antibiotics employed to treat drug-resistant organisms may also be associated with severe side effects, including hepatotoxicity. Thus, novel strategies to treat drug-resistant tuberculosis are urgently needed wherein combining antibiotics with immunomodulators may yield more favorable outcomes [10–13].

Clofazimine (CFZ) shows treatment efficacy against *M. tuberculosis* and *M. leprae* infection and is employed to treat lepromatous leprosy [14]. Furthermore, we recently reported that CFZ promotes the generation of central memory T (T_{CM}) cells, which facilitated the vaccine efficacy of Bacillus Calmette Guérin (BGC) [15]. This finding is in agreement with previous reports that *M. tuberculosis* induces T effector memory (T_{EM}) cells in the lungs, which exacerbates tuberculosis pathogenesis, whereas T_{CM} cells may ameliorate tuberculosis disease [16]. Therefore, CFZ may provide advantages over other tuberculosis drugs for the treatment of drug-resistant tuberculosis.

Tuberculosis pathogenesis consists of infection and inflammation, and successful treatment requires not only elimination of *M. tuberculosis* organisms but also amelioration of inflammatory responses. Consistent with this notion, addition of steroids to conventional antibiotic therapy displays improved treatment efficacy [17] and hence anti-inflammatory agents may prove beneficial during tuberculosis treatment. Steroids have anti-inflammatory properties but are also associated with serious side effects and potential posttherapy recurrence of the disease [18]. Rapamycin (RAPA) is an anti-inflammatory

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agent that has also been shown to kill intracellular *M. tuberculosis* organisms [19] by inducing autophagy, an added benefit over steroids. RAPA also induces immunostimulatory effects by generating memory CD8⁺ T cells. Furthermore, RAPA increases the number of memory precursors during the expansion phase and accelerates memory T-cell differentiation during the contraction phase [20]. Moreover, RAPA increases the survival of antigen-specific CD8⁺ T cells [20,21] by switching cellular metabolism from glycolysis to fatty acid oxidation [22]. RAPA not only increases the magnitude of the CD8⁺ T-cell response but also the functional qualities of antigen-specific CD8⁺ T_{CM} cells, imparting long-lived protective immunity [23].

Here we show that a combination of RAPA and CFZ completely eliminates MDR and XDR *M. tuberculosis* strains in mice. This combined therapy successfully ameliorated inflammatory lesions, induced host protective immune responses, and favored T_{CM} over T_{EM} subsets. CFZ-RAPA cotreatment also downregulated the dormancy-associated genes of drug-resistant *M. tuberculosis* in human macrophages. The current regimen to treat patients with MDR tuberculosis includes a combination of 4 drugs, moxifloxacin, kanamycin, linezolid, and ethionamide, and is a very lengthy treatment and results in serious side effects. However, the present study includes only 1 antibiotic, CFZ along with the mechanistic target of rapamycin (mTOR) inhibitor, RAPA for treating both MDR and XDR tuberculosis. Therefore, this combination shows promising prospect over current regimens for treating drug-resistant *M. tuberculosis* infections.

METHODS

Ethics Statement

Animal experiments were performed according to the guidelines approved by the Institutional Animal Ethics Committee of International Centre for Genetic Engineering and Biotechnology (ICGEB), New Delhi, India, and the Department of Biotechnology (DBT), Government of India, guidelines. All mice used for experiments were ethically sacrificed by asphyxiation in carbon dioxide according to institutional and DBT, Government of India, regulations.

Mice

Female Balb/c and C57BL/6 (6–8 weeks of age) mice were provided by ICGEB, New Delhi, India. For the adoptive transfer experiments, we used female C57BL/6 mice (6–8 weeks old), because Thy1.1 and Thy1.2 were available in C57BL/6 background.

M. tuberculosis Infection of Mice and Estimation of Colony Forming Units

Drug-resistant strains of *M. tuberculosis* (multidrug resistant, Jal 1934, Jal 2287, Jal 2261, BND 320, and Jal 2261, and

extensively drug resistant, MYC 431) were a kind gift from the JALMA Institute, Agra, India [24]. All cultures were grown in Middlebrook 7H9 (Difco) medium supplemented with 10% ovalbumin, dextrose, and catalase (OADC; Difco), 0.05% Tween 80, and 0.2% glycerol till mid-log phase. Aliquots of the cultures in 20% glycerol were preserved at –80°C and these cryopreserved stocks were used for infections.

Mice were infected with drug-resistant strains of *M. tuberculosis* (Jal 2261 and MYC 431) via the aerosol route using a Madison aerosol chamber (University of Wisconsin, Madison, WI) with its nebulizer precalibrated to deposit a total of approximately 110 colony forming units (CFU) to the lungs of each mouse as previously described [25,26]. Five randomly selected mice were sacrificed at various time points and organs were harvested, homogenized in 0.2 µm filtered phosphate-buffered saline (PBS) containing 0.05% Tween 80, plated onto Middlebrook 7H11 (Difco) plates containing 10% OADC (Difco), and incubated at 37°C for 21–28 days. Bacterial colonies were counted and CFU were estimated as per dilution.

Drug Administration

CFZ (Sigma) 5 mg/kg, in 100 µL of PBS containing 5% dimethyl sulfoxide (DMSO) volume/volume, was administered intraperitoneally to the mice every day, whereas control mice were given vehicle only. A relatively low dose of 80 µg/kg of RAPA (Sigma) was used as a higher dose has been reported to suppress T cells during the expansion phase of the CD8⁺ T-cell response [27,28]. Moxifloxacin, kanamycin, linezolid, and ethionamide were given at a dose of 50 mg/kg, 5 days a week.

Flow Cytometry: Surface and Intracellular Staining

Spleens and lungs were isolated from mice and macerated by frosted slides in ice cold RPMI 1640 (Gibco, Invitrogen) containing 10% fetal bovine serum (FBS) to prepare a single-cell suspension. The splenocytes were cultured after RBC lysis buffer 1 × 10⁶ cells were cultured per well in 12-well plates (Nunc) and activated with H37RV complete soluble antigen overnight, and 10 µg/mL brefeldin A (eBiosciences) was added for 6 hours for intracellular staining. Cells were washed twice with PBS and stained with antibodies directed against surface markers, followed by fixation and permeabilization, and stained with anticytokine antibodies. Fluorescence intensity of fluorochrome-labelled cells was measured by flow cytometry (FACS Canto II; BD Biosciences). FACS Diva was used for acquiring the cells and data analysis were performed by Flowjo (Treestar).

Antibodies and Reagents

We used the following antibodies: anti-CD3(clone:17A2)-PerCP-Cy5, -APC, -PE Cy7, -Pacific Blue; anti-CD4(clone: GK1.5, RM4-5)-FITC, -PerCP-Cy5, -APC; anti-CD8(clone:53-6.7)-FITC, -APC-Cy7, -PerCP-Cy5, -APC; anti-CD44(clone:

IM7)-APC; anti-CD62L(clone:MEL-14)-PE; anti-CD25(clone: 3C7)-PE, -APC; anti-FOXP3(clone:MF14/150D)-APC; anti-IFN- γ (clone:XMG1.2)-APC; anti-IL-4(clone:11B11)-PE; anti-IL-17(clone:TC11-18H10.1)-PE, -PE-Cy7; anti-TNF- α (clone: MP6-X22)-PE, -APC, -FITC; anti-CD69(clone: H1.2F3)-PE, -FITC; and anti-PD-1(clone:29F.1A12)-PE (all from Biolegend).

Preparation of Peritoneal Macrophages

C57BL/6 mice were injected intraperitoneally with 2 mL of 4% thioglycollate (Brewer modified, BBL, BD). Five days later, peritoneal exudate cells were isolated from the peritoneal cavity. Cells were cultured overnight at 37°C, 5% CO₂ and washed with RPMI-10% FBS to remove nonadherent cells. Adherent monolayer cells were used as peritoneal macrophages. These peritoneal macrophages were infected with H37Rv, MDR, and XDR strains of *M. tuberculosis* at multiplicity of infection of 1:10 for 4 hours followed by washing with fresh medium and incubation in 5% CO₂ (37°C) in complete RPMI-1640 medium (Invitrogen).

qPCR of RNA from Infected and Treated Cells

RNA was isolated from the infected and treated cells by using Trizol. Bacterial cDNA was prepared using iScript cDNA synthesis kit (Bio-Rad). Real-time quantitative polymerase chain reaction (qPCR) was performed in 10 μ L reaction volume using SYBR Green as per manufacturer's instructions and CFX96 Real Time PCR Detection System (Bio-Rad). All reactions were normalized to 16S rRNA for *M. tuberculosis*.

Statistical Analysis

All mice experiments were performed twice with 5 mice per group per time point ($n = 5$). Significant differences between the groups were determined by *t* test and 1-way analysis of variance followed by multiple Tukey test. A value of $P \leq .05$ was considered statistically significant.

RESULTS

Simultaneous Treatment With CFZ and RAPA Restricts the Survival of MDR and XDR Strains of *M. tuberculosis* in Mice

C57BL/6 mice were infected with MDR or XDR strains of *M. tuberculosis* and rested for an additional 15 days. Figure 1A shows the infection model. Cotreatment with CFZ and RAPA dramatically reduced the bacterial burden in the lungs of mice 60 days postinfection (Figure 1B and 1C) compared with other groups. No detectable CFU was observed at 100 days postinfection in the lungs of mice cotreated with CFZ and RAPA, while few mice in the other treatment groups failed to survive. The CFZ and RAPA combination was also tested ex vivo in peritoneal macrophages against a series of drug-sensitive and drug-resistant strains of *M. tuberculosis*. In most of the cases, addition of RAPA further decreased the bacterial survival as compared to CFZ treatment alone (Supplementary Figure 1A).

To compare our treatment strategy with the existing therapy, we treated MDR-infected mice with a combination of 4 drugs (moxifloxacin, kanamycin, linezolid, and ethionamide) with or without RAPA [29] (Supplementary Figure 1B). Combinatorial treatment of 4 drugs along with RAPA led to a 3-log reduction in the bacterial load; however, the effect achieved was much

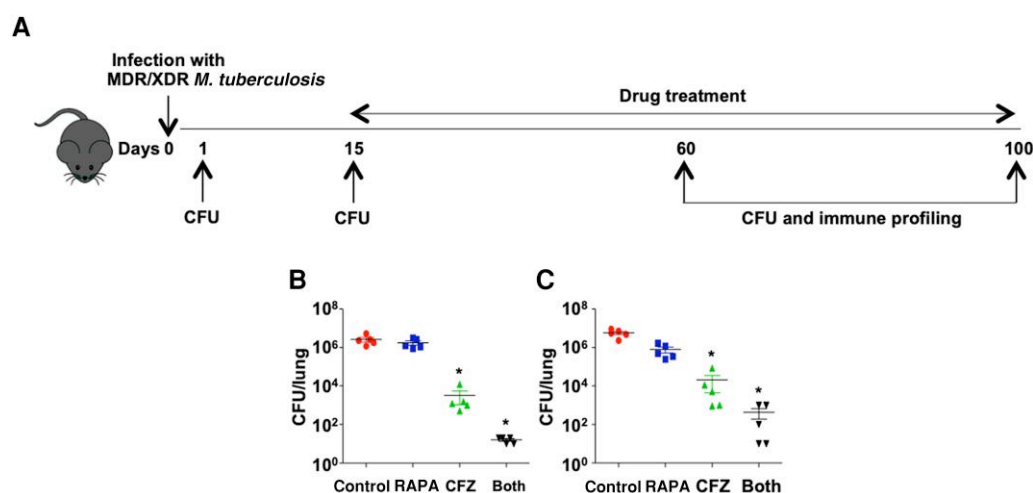


Figure 1. Combined treatment with CFZ and RAPA protects mice against tuberculosis. A, Schematic diagram to show the groups of naive C57BL/6 mice challenged with MDR (Jal-2261) or XDR (MYC-431) strains of *Mycobacterium tuberculosis* via the aerosol route with a low-dose inoculum of approximately 110 CFU/mice. After 2 weeks, these mice were treated with CFZ and RAPA for the next 85 days. Mice were sacrificed at various time points and lungs were harvested for the estimation of bacterial burden; CFU from lung homogenates of the different groups of (B) MDR-infected and (C) XDR-infected mice. Data are representative of 2 independent experiments ($n = 5$ mice/group). * $P \leq .05$. Line and Error bar means $\pm$ SD. Abbreviations: CFU, colony-forming unit; CFZ, clofazimine; MDR, multidrug resistant; RAPA, rapamycin; XDR, extensively drug resistant.

less compared to the CFZ-RAPA cotreatment, which decreased the bacterial load by 5 logs (Supplementary Figure 1B and 1F). These data suggest that combination therapy with CFZ and RAPA could be a better and successful alternative against drug-resistant strains of *M. tuberculosis*.

Combination Therapy Enhances Immunity Against *M. tuberculosis*

To investigate the impact of combined therapy with CFZ and RAPA on host immune responses, we profiled the T-cell

population in the lungs and the spleen of infected animals. The combination therapy not only enhanced the prevalence of CD4⁺ and CD8⁺ T cells in the lungs (Supplementary Figure 2A) and the spleen of mice infected with MDR and XDR strains of *M. tuberculosis* (Figure 2A and 2B), but also induced T-cell activation in the spleen (Figure 2C) and the lungs (Supplementary Figure 2B) of infected animals. The numbers of lung CD4⁺ and CD8⁺ T cells in the RAPA group were significantly higher; however, there was no significant difference in

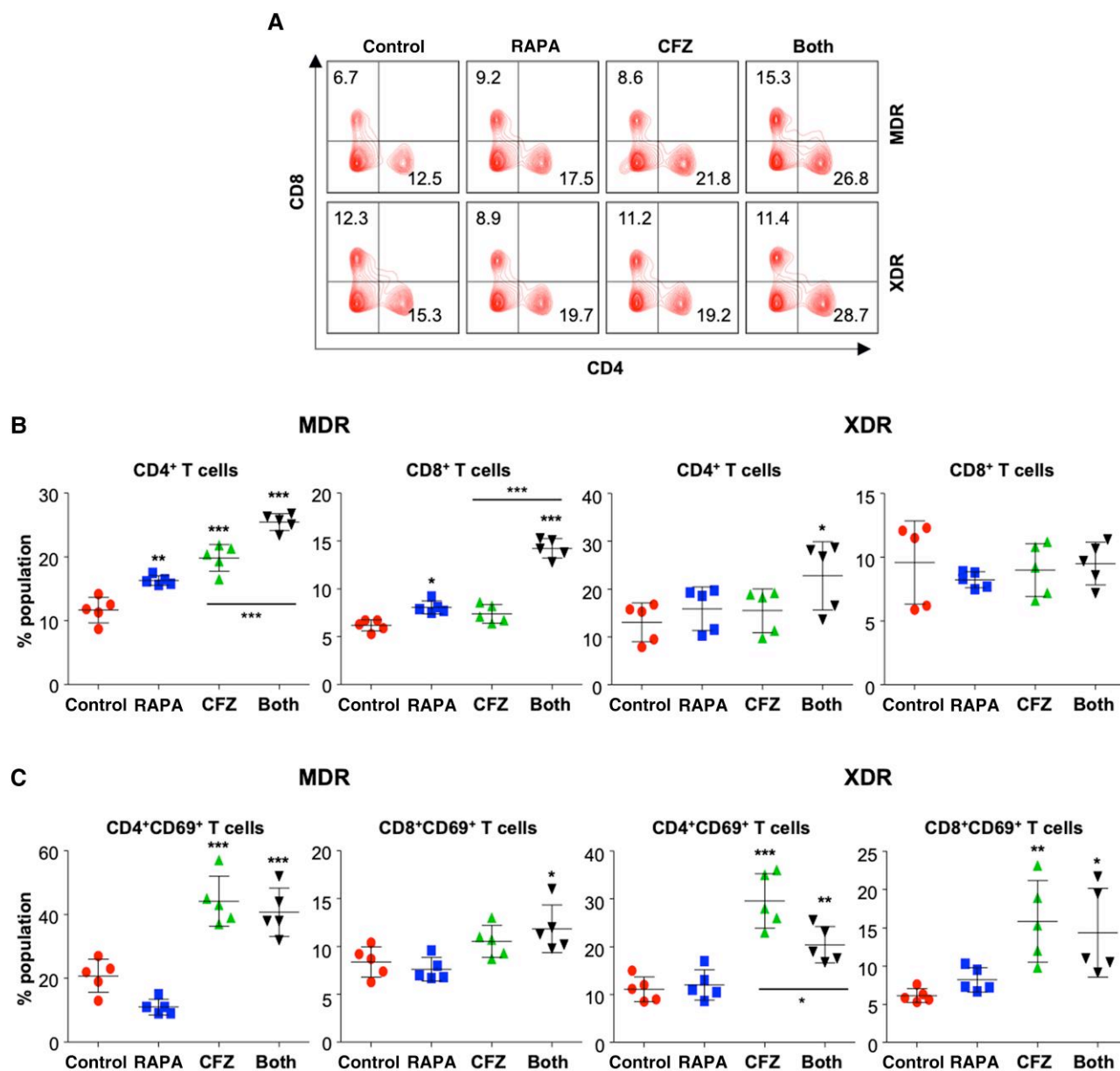


Figure 2. Combined therapy with CFZ and RAPA induces the activation of T lymphocytes. *A*, Representative dot plots and (*B*) the percentage of CD4⁺ and CD8⁺ T cells in the spleen of different groups of mice infected with MDR or XDR strains of *Mycobacterium tuberculosis*. *C*, The percentage of CD69-expressing CD4⁺ and CD8⁺ T cells in the spleen of infected mice. Data shown are representative of 2 independent experiments with 5 mice in each group. Statistical analysis was performed by 1-way ANOVA followed by Tukey *t* test. * *P* < .05, ** *P* < .005, *** *P* < .0005 (control vs other groups). Line and Error bar means ± SD. Abbreviations: CFZ, clofazimine; MDR, multidrug resistant; RAPA, rapamycin; XDR, extensively drug resistant.

their activation status, indicating that CFZ has a superior ability to drive T-cell activation (Figure 2A–2C). Next, we analyzed the expression of memory cell markers CD62L and CD44. We observed that the combination therapy enriched the CD4⁺/CD8⁺CD62L^{hi}CD44^{hi} T_{CM} cell pool (CD4⁺/CD8⁺CD62L^{hi}CD44^{hi}) while decreasing the T_{EM} cell pool (CD4⁺/CD8⁺CD62L^{lo}CD44^{hi}) in the spleen (Figure 3A–3C) and the lungs (Supplementary Figure 3) of

infected mice as compared with the other experimental groups. To understand the direct effect of CFZ and RAPA on T-cell differentiation, T cells were isolated from *M. tuberculosis*-infected mice, ex vivo stimulated with *M. tuberculosis* complete soluble antigen, and treated with CFZ and/or RAPA for 48 hours, followed by immune profiling. We observed that CFZ and RAPA combination therapy significantly induced the differentiation

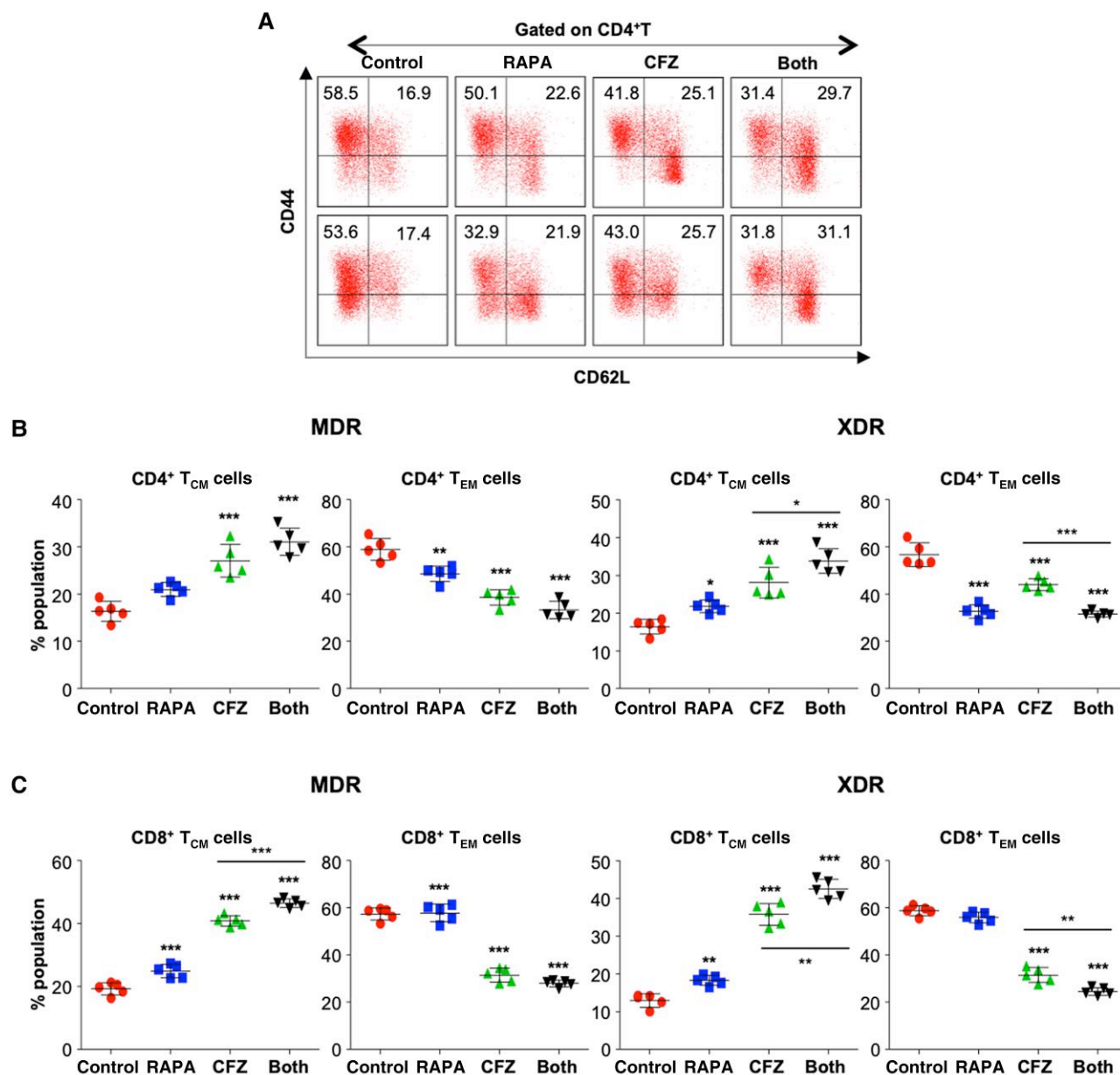


Figure 3. CFZ and RAPA cotreatment induces central memory T-cell responses. A, Dot plots and (B) the percentage of memory immune responses (T_{CM}-CD4⁺CD62L^{hi}CD44^{hi}, T_{EM}-CD4⁺CD62L^{lo}CD44^{hi}) of CD4⁺ T cells isolated from the spleen of infected and treated mice. C, The percentage of CD8⁺ memory immune responses in the spleen of mice infected with MDR and XDR tuberculosis. Data shown are representative of 2 independent experiments with 5 mice in each group. Statistical analysis was performed by 1-way ANOVA followed by Tukey *t* test. * *P* < .05, ** *P* < .005, *** *P* < .0005 (control vs other groups). Line and Error bar means ± SD. Abbreviations: CFZ, clofazimine; MDR, multidrug resistant; RAPA, rapamycin; XDR, extensively drug resistant.

of naive T cells (T_N) into stem cell memory (T_{SCM}), T_{EM} , and T_{CM} T-cell subsets (Supplementary Figure 4), which is in line with our previous observation that CFZ, a potassium ion blocker, significantly induces central memory T-cell responses [15]. Interestingly, RAPA cotreatment enhanced the T-cell differentiation-driving potential of CFZ. Taken together, we concluded that combined therapy with CFZ and RAPA could induce a faster and stronger recall response upon subsequent infection, owing to their large pool of T_{CM} cells in the spleen and the lungs of infected animals.

Combination Therapy Regulates PD-1 Expression in T_{CM} and T_{EM} Cell Subsets

The programmed death-1 (PD-1; CD279) receptor is a negative regulator of immune responses and is predominantly expressed by exhausted or otherwise dysfunctional T cells [30] as it inhibits T-cell activation and cytokine production [31]. Moreover, PD-1 immune checkpoint blockade results in disease resistance [32]. We found that the combined therapy with CFZ and RAPA reduced the level of PD-1 on both the T_{CM} and T_{EM} subsets of $CD4^+$ T and $CD8^+$ T cells in the spleen (Figure 4A) and the lungs (Supplementary Figure 5A) of mice infected with MDR or XDR strains of *M. tuberculosis*, compared with the groups

receiving no or single treatment only. Taken together, CFZ and combination therapy with CFZ-RAPA inhibits the expression of the immune inhibitory PD-1 molecule, thus allowing the host to induce protective immune responses. Further analysis revealed that the CFZ alone and the combined therapy elicited T_{CM} cells producing cytokines associated with polyfunctional $CD4^+$ and $CD8^+$ T cells ($CD62L^{hi}CD44^{hi}IFN-\gamma^+TNF-\alpha^+IL-2^+$) in the spleen (Figure 4B) and the lungs of infected animals (Supplementary Figure 5B), which are thought to be protective precursors against tuberculosis [33,34].

Clofazimine and Rapamycin Cotreatment Reduces the Rate of Latent Tuberculosis Reactivation

Because the combination was shown to induce central memory T-cell responses, which may protect during a secondary infection, we performed reactivation studies in the mice model of tuberculosis (Figure 5A). Interestingly, reactivation of the bacilli was observed in the lungs of only 3 mice out of 10 with CFZ and RAPA cotreatment as compared to a 60% relapse rate in the mice group treated with INH and Rifampicin (Figure 5B). Moreover, the bacillary load was significantly less in the CFZ-RAPA-treated group (Figure 5B).

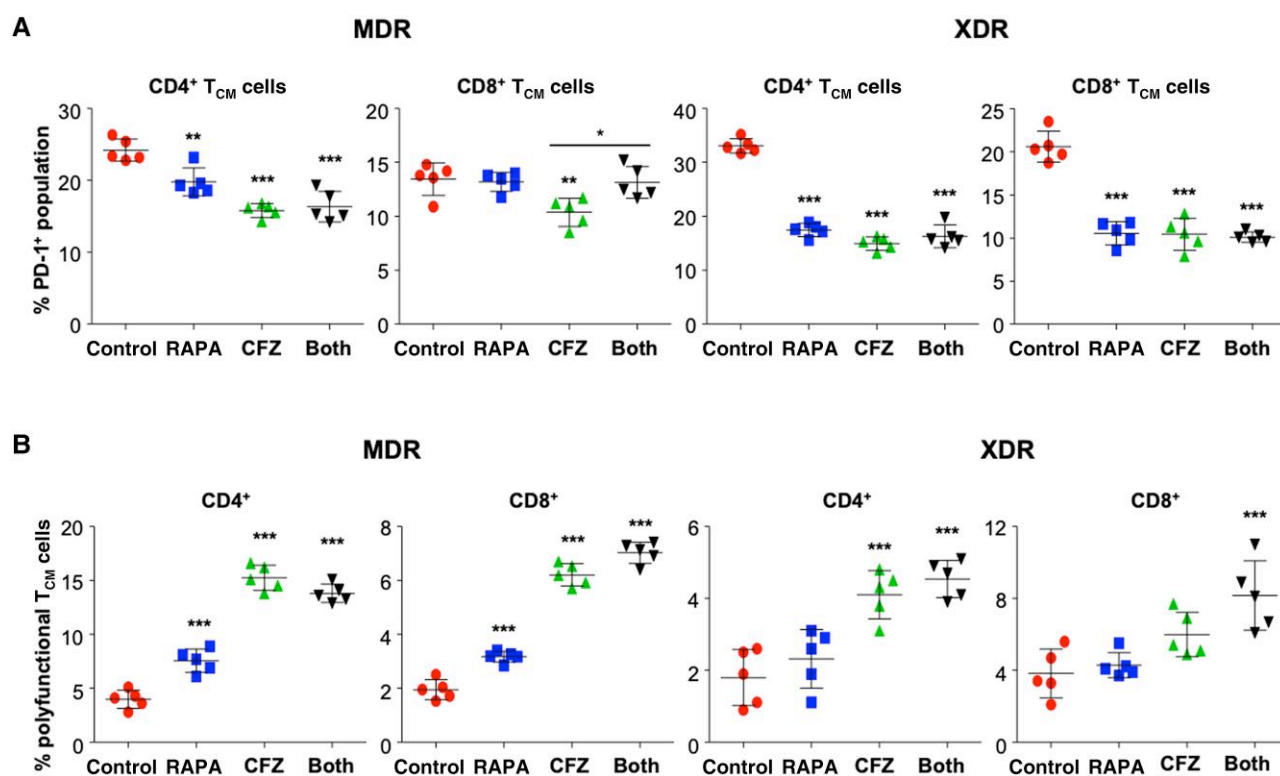


Figure 4. T_{CM} cells from cotreated mice display reduced PD-1 expression and heightened polyfunctional cytokine responses. *A*, Percentage of $CD4^+$ and $CD8^+$ T_{CM} cells expressing the PD-1 receptor in the spleen of infected animals. *B*, Profiling of the polyfunctional cytokine responses ($IFN-\gamma^+IL-2^+TNF-\alpha^+$) from the T_{CM} subset of the splenic $CD4^+$ and $CD8^+$ T cells. Data shown are representative of 2 independent experiments with 5 mice in each group. Statistical analysis was performed by 1-way ANOVA followed by Tukey *t* test. * $P < .05$, ** $P < .005$, *** $P < .0005$ (control vs other groups). Line and Error bar means $\pm$ SD. Abbreviations: CFZ, clofazimine; MDR, multidrug resistant; PD-1, programmed death-1; RAPA, rapamycin; XDR, extensively drug resistant.

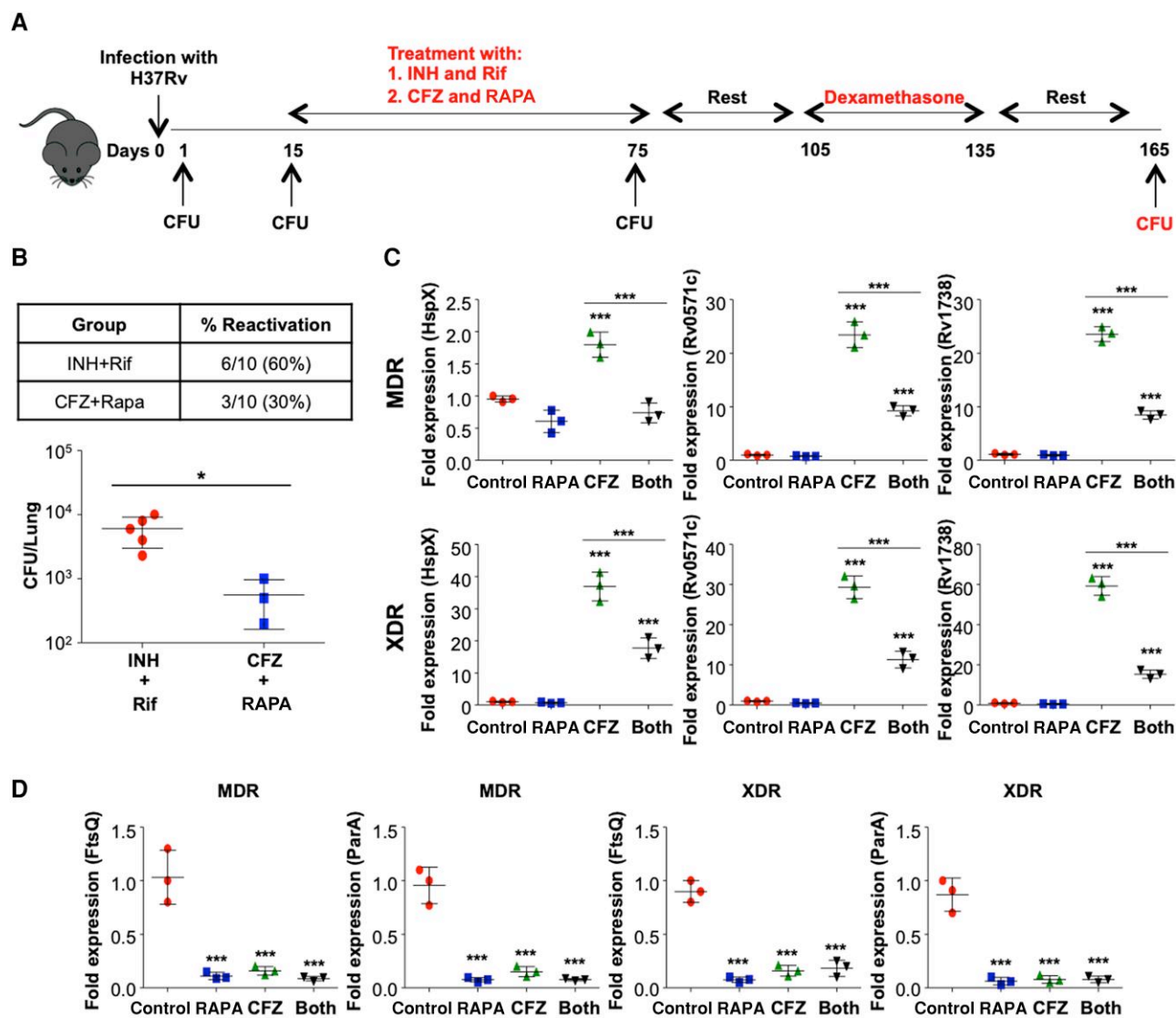


Figure 5. CFZ and RAPA cotreatment reduces the tuberculosis reactivation rate in mice. **A**, Schematic representation of the experimental design. **B**, The number of mice showing relapse of the disease and the bacterial load in the lungs of reactivated mice. THP1 cells were infected with MDR and XDR strains of *Mycobacterium tuberculosis* at a multiplicity of infection of 1:10 followed by treatment with CFZ and RAPA. **C**, Relative expression of dormancy genes of intracellular *M. tuberculosis* treated with CFZ and RAPA at 48 hours postinfection as compared with the untreated bacilli. **D**, Fold change in the expression of replicative genes in intracellular *M. tuberculosis*. 16S gene was taken as internal control. **A** and **B**, The results are representative of 1 independent experiment with 10 mice in each group. * $P < .05$. **C** and **D**, Data are representative of 3 independent experiments performed in triplicates ($n = 3$, $\pm$ SD). ** $P < .005$, *** $P < .0005$. Line and Error bar means $\pm$ SD. Abbreviations: CFU, colony-forming unit; CFZ, clofazimine; INH, isoniazid; MDR, multidrug resistant; RAPA, rapamycin; Rif, Rifampicin; XDR, extensively drug resistant.

RAPA Cotreatment Reduces the Expression of CFZ-Induced Latency Genes in the MDR and XDR Strains of Tuberculosis in the Human Macrophages

Antibiotic treatment is known to induce stringent response in the bacteria, which may lead to the generation of persistent/latent populations [35,36]. The reactivation results encouraged us to investigate the expression of dormancy-related genes after cotreatment with CFZ and RAPA. To this end, THP1, human monocyte-like cells, were infected with MDR and XDR *M. tuberculosis* followed by treatment with RAPA and CFZ. At 48 hours postinfection, cells were lysed and bacterial RNA was isolated followed by real-time PCR to analyze the expression of replicative and latency genes [37]. Interestingly, we

observed that CFZ treatment significantly induced the expression of dormancy-related genes while RAPA cotreatment led to a marked downregulation of these genes (Figure 5C) with no significant difference in the replicative genes (Figure 5D). Taken together, these data strongly indicate that the bacteria fail to establish dormancy in the case of CFZ-RAPA cotreatment in human macrophages.

Combination Therapy Promotes *M. tuberculosis*-Specific, Proinflammatory Cytokine Responses

To further understand the *M. tuberculosis* antigen-specific responses we estimated the levels of key cytokines secreted by

T cells from the spleen and the lung of infected animals in response to ex vivo stimulation with complete soluble antigen. Antigen-specific CD4⁺ T cells from the spleen of infected mice treated with combination therapy produced markedly increased levels of the proinflammatory cytokines interferon- γ (IFN- γ) and interleukin 17 (IL-17) (Figure 6A) and reduced levels of the anti-inflammatory cytokine IL-4 and regulatory T (Treg) cell responses (Figure 6B). Similar response was observed in the splenic CD8⁺ T cells (Figure 6C and 6D). We observed increased prevalence of IFN- γ , IL-17, and IL-4-producing CD4⁺ and CD8⁺ T cells in most of the cases with no significant difference in Treg cells in the lungs (Supplementary Figure 6). Our findings suggested that CFZ alone and CFZ and RAPA cotreatment enhance antigen-specific protective Th1 and Th17 responses, and to an extent inhibit Th2 and Treg cells. Therefore, these data suggest that combination therapy promotes host protective responses against *M. tuberculosis*.

T Cells From Infected Mice Treated With Combination Therapy Confer Host Protective Immunity Against *M. tuberculosis* Infection Upon Adoptive Transfer

To further investigate the functional quality and specificity of T cells, we performed adoptive transfer experiments. Congenic wild-type Thy1.2⁺ mice were infected with MDR *M. tuberculosis*, treated with CFZ and RAPA (5 mg/kg) for 30 days, and then rested for 60 days. CD4⁺ and CD8⁺ T cells (5×10^6) isolated from these animals were then adoptively transferred into γ -irradiated (sublethal dose of 800 rads/mice) Thy1.1⁺ congenic animals. These animals were subsequently challenged with a low dose of MDR *M. tuberculosis* by the aerosol route. Twenty-one days after infection, spleens were isolated for immune profiling and lungs were harvested to determine the bacterial load (Figure 7A). Mice receiving cells from CFZ and RAPA cotreated mice maintained increased levels of IFN- γ and IL-17-producing CD4⁺ and CD8⁺ T cells (Figure 7B), which led to enhanced bacterial killing (Figure 7C), reiterating that CFZ-RAPA cotreatment induced protective *M. tuberculosis*-specific immune responses.

DISCUSSION

Novel strategies that consider both host immune responses and infection dynamics are urgently needed to combat drug-resistant tuberculosis. It is now increasingly clear that modulation of host immune responses is essential for successful elimination of *M. tuberculosis* from the host. Hence, adjunct therapy of antibiotics with an immune modulator is an attractive strategy for treatment of drug-resistant tuberculosis. Th1 and Th17 cells play a critical role in host resistance to *M. tuberculosis* infection [38–40]. CFZ is an approved drug for the treatment of leprosy, and also exhibits efficacy against *M. tuberculosis* infection [15, 16]. CFZ has not yet been extensively

employed for the treatment of regular tuberculosis and is currently reserved for treating MDR/XDR infections in humans. *M. tuberculosis* has not yet acquired resistance to CFZ. CFZ is also an inhibitor of the Kv1.3⁺ potassium channel, which is predominantly expressed by effector T cells, and thus reduces inflammation [15]. It also favors the generation of T_{CM} over T_{EM} cells [15].

The available tuberculosis antibiotics are ineffective in clearing nonreplicating bacteria, some of which are sequestered in distinct tissues [41,42]. Elimination of such bacteria requires immune modulation. We have previously shown that induction of autophagy by RAPA successfully eliminates *M. tuberculosis* from mesenchymal stem cells and macrophages [43]. Additionally, it has been shown that inhibition of autophagy contributes to the capacity of virulent *M. tuberculosis* organisms to survive within macrophages [44–46]. RAPA is also known to influence the generation of memory CD8⁺ T cells through inhibition of mTOR [20–23].

With this background, we designed a novel treatment regime against MDR and XDR strains of *M. tuberculosis* wherein mice were cotreated with defined and nontoxic doses of CFZ and RAPA and bacillary load was monitored over time. Interestingly, we found that RAPA and CFZ cotreatment significantly reduced the bacterial burden and improved the gross pathology of mice infected with MDR and XDR tuberculosis (Figure 1). There was a marked increase in the percentage of polyfunctional CD4⁺/CD8⁺ T_{CM} cells secreting IFN- γ , TNF- α , and IL-2 cytokines in the spleen and the lungs of mice treated with CFZ alone and also in the cotreated group (Figure 4).

Tuberculosis pathogenesis is mainly controlled by induction of proinflammatory Th1 cytokines such as IFN- γ , IL-12, and TNF- α , which limit bacterial replication in macrophages [47]. These cytokines, along with IL-17, also play an important role in granuloma formation [48]. On the other hand, Th2 cytokines IL-4 and IL-10 favor tuberculosis pathogenesis, disease progression, and reactivation [47]. In short, the balance between Th1 and Th2 cells is critical for antituberculosis immune response. Interestingly, CFZ alone as well as cotreatment with CFZ and RAPA induced an effective immune response in favor of the host with increased secretion of protective cytokines (IFN- γ and IL-17) along with the decreased levels of anti-inflammatory cytokines (IL-4), which was reflected in decreased survival of drug-resistant bacteria in treated mice (Figure 6).

Establishment of a long-lived T-cell memory response (T_{EM} and T_{CM}) is a critical prerequisite for obtaining protection against reactivation and reinfection with tuberculosis [49]. Later studies provided evidence for T_{CM} being the major player in providing protection against tuberculosis [50]. Consistent with its ability to induce T_{CM}, CFZ along with RAPA induced a higher number of CD62L^{hi} cells with a concomitant decrease

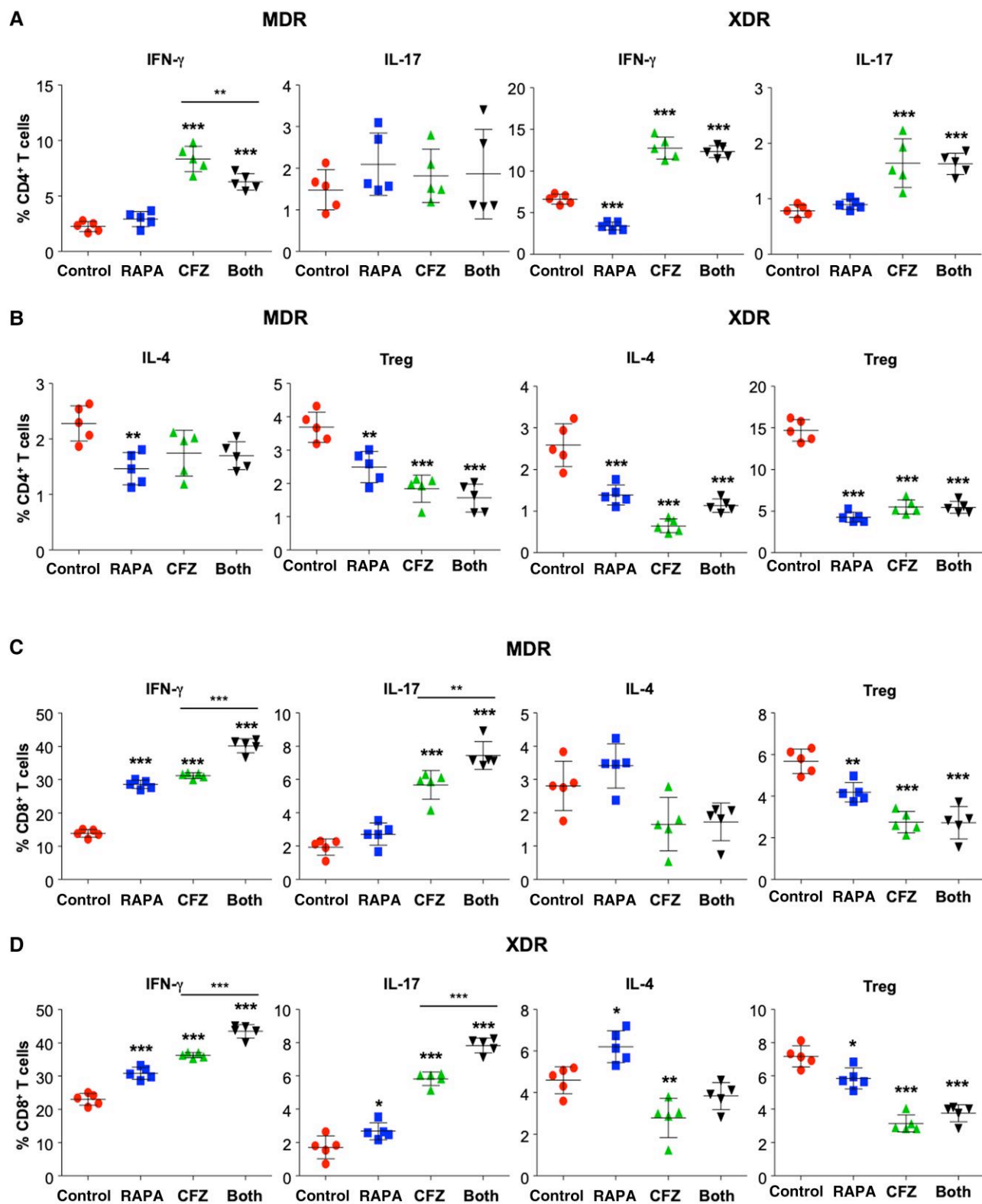


Figure 6. CFZ and RAPA cotreatment induces proinflammatory responses while reducing anti-inflammatory and inhibitory immune responses. **A**, Profiling of IFN- γ and IL-17 cytokines in the CD4⁺ T cells from the spleen of mice infected with MDR and XDR strains of *Mycobacterium tuberculosis*. **B**, Percentage of IL-4⁺CD4⁺ T cells and CD4⁺CD25⁺FOXP3⁺ cells in the spleen of mice infected with MDR and XDR strains of *M. tuberculosis*. **C** and **D**, Percentage of CD8⁺ T cells expressing IFN- γ , IL-17, IL-4, and CD25-FOXP3 in the spleen of mice infected with MDR or XDR strains of *M. tuberculosis*. Data are representative of 2 independent experiments with 5 mice in each group. Statistical analysis was performed by 1-way ANOVA followed by Tukey *t* test. * $P < .05$, ** $P < .005$, *** $P < .0005$ (control vs other groups). Line and Error bar means $\pm$ SD. Abbreviations: CFU, colony-forming unit; CFZ, clofazimine; IFN- γ , interferon- γ ; IL, interleukin; MDR, multidrug resistant; RAPA, rapamycin; Treg, regulatory T cell; XDR, extensively drug resistant.

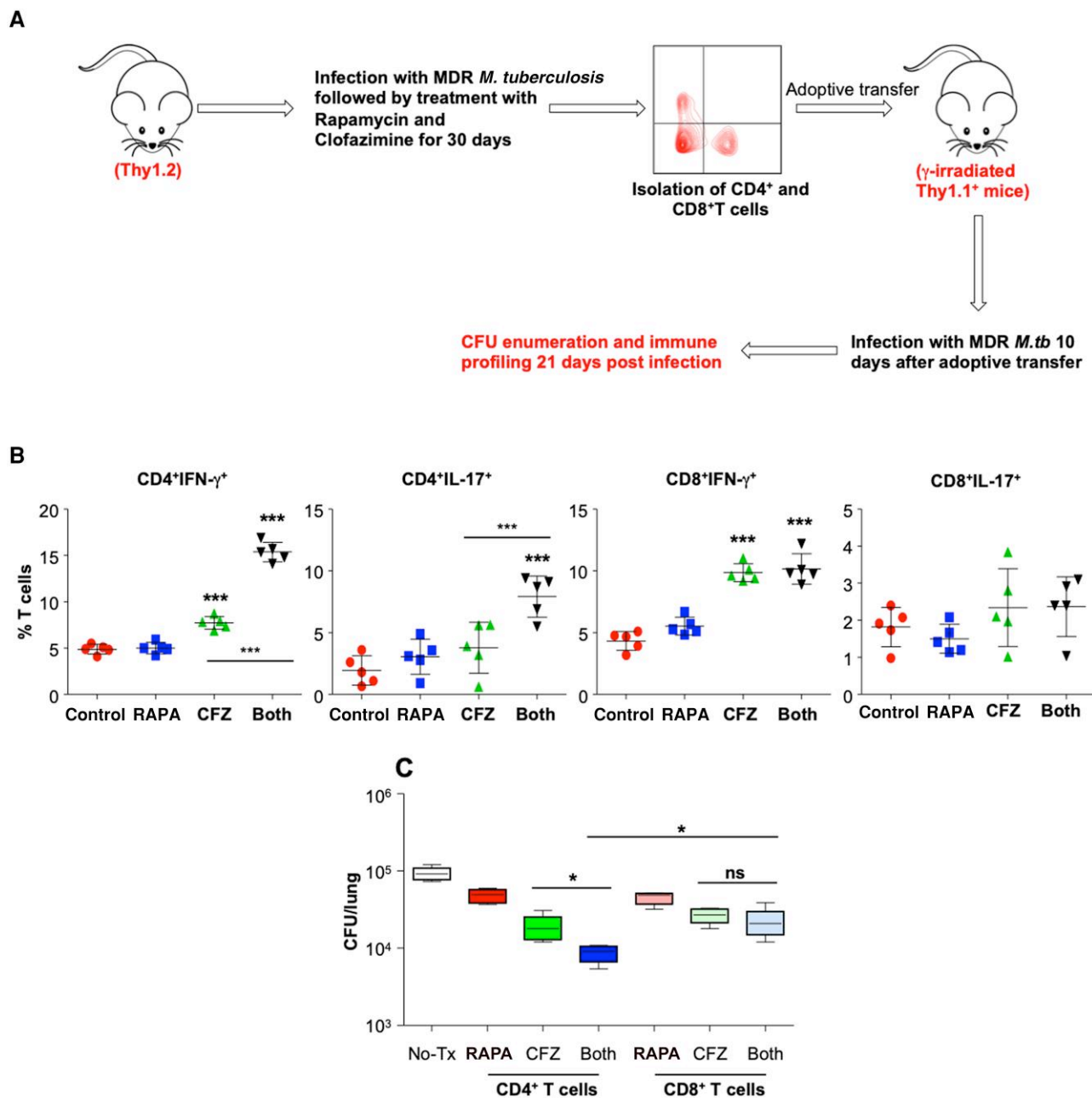


Figure 7. CD4⁺ T cells from infected mice treated with CFZ and RAPA confer improved protection against tuberculosis. **A**, Experimental layout: Thy1.1⁺ mice were γ-irradiated (8 rads/s for 100 s), rested for 1 day, and divided into 7 different groups. The Thy1.1⁺ control group that did not receive any adoptively transferred T cells served as a negative control. CD4⁺ and CD8⁺ T cells were enriched using anti-CD4/anti-CD8 beads from the lymph nodes of Thy1.2⁺ animals that had been previously infected with MDR and treated with CFZ and RAPA for 30 days and then rested for an additional 30 days. These cells were then adoptively transferred into the irradiated recipient mice (5×10^6 cells per mouse). Four days after adoptive transfer, all the groups of irradiated Thy1.1⁺ mice were challenged with MDR *Mycobacterium tuberculosis* through the aerosol route. The mice were euthanized for profiling of intracellular cytokines and CFU enumeration at 21 days after aerosol challenge. **B**, Splenocytes were challenged with MDR complete soluble antigen ex vivo and T cells were then stained for expression of the intracellular cytokines IFN-γ and IL-17 from the CD4⁺ and CD8⁺ T cells. **C**, Bacterial burden in the lungs of adoptively transferred and infected animals. The results shown are representative of 1 independent experiment with 5 mice in each group. Statistical analysis was performed by 1-way ANOVA followed by Tukey *t* test. * *P* < .05, ** *P* < .005, *** *P* < .0005 (control vs other groups). *t* test was performed in C. Line and Error bar means ± SD. Abbreviations: CFU, colony-forming unit; CFZ, clofazimine; IFN-γ, interferon-γ; IL, interleukin; MDR, multidrug resistant; RAPA, rapamycin; Tx, treatment; XDR, extensively drug resistant.

in CD62L^{lo} cells (Figure 3). Interestingly, RAPA cotreatment significantly enhanced the ability of CFZ to induce T_N differentiation into different T-cell memory subsets (Supplementary Figure 4). Furthermore, CFZ-induced expression of *M.*

tuberculosis latency genes was significantly controlled upon RAPA cotreatment (Figure 5). This feature imparts an added advantage to this drug combination by reducing the danger of reactivation and reinfection of the disease in cotreated

mice (Figure 5). Adoptive transfer experiments further indicated the combined ability of CFZ and RAPA to induce enhanced killing potential in T cells (Figure 7).

Overall, we demonstrate that CFZ and RAPA complement each other by modulating critical mechanisms of the host and the bacilli for better treatment outcome. Lastly, both of these drugs are approved for use in human subjects and may be useful in developing an immunostimulatory drug regimen for treatment of infections caused by drug-resistant *M. tuberculosis* strains and other microbial organisms. However, these results should be replicated in nonhuman primates to clearly demonstrate the potential of CFZ-RAPA combination against drug-resistant tuberculosis.

Supplementary Data

Supplementary materials are available at *The Journal of Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copy-edited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

Author contribution. D. K. S., V. P. D., and G. D. conceived the hypothesis. D. K. S., A. B., V. P. D., and G. D. designed the experiments. D. K. S., A. B., and V. P. D. performed all the experiments and analyzed data. A. B., I. P., A. S., B. M., and S. C. performed ex vivo experiments. S. C. helped in animal experiments. D. K. S., A. B., V. P. D., and G. D. wrote the manuscript. V. P. D. and G. D. supervised the experiments.

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